

Portfolio Analytics Dashboard

Interactive analytics and decision-support application demonstrated through a mutual fund portfolio use case

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Project Overview

This project demonstrates an end-to-end approach to transforming structured transactional data into an interactive, high-level analytics dashboard. Mutual funds are used as the demonstration domain; the underlying framework is intended to be reusable across finance, workforce, sales, operations and other business analytics scenarios.

Core capabilities

- Portfolio and fund-level performance analysis
- Holdings, profitability and return analysis
- XIRR-based investment performance analysis
- Asset, market-cap and AMC allocation views
- Interactive tables, filtering, sorting and visual analytics
- Dashboard-ready data processing from Excel transaction data

1. Business Purpose

The dashboard converts transaction-level information into a consolidated view that helps users understand where the portfolio stands, how individual funds are performing and where exposure or performance requires attention.

The mutual fund use case is deliberately concrete and easy to demonstrate, but the larger objective is the capability to build reusable business dashboards around structured data.

2. Data-to-Insight Mechanism

- Excel transaction data is loaded and structured for analysis.
- Transaction history is processed to derive holdings, investment and redemption information.
- Financial and portfolio metrics are calculated at fund and portfolio levels.
- Analytical views are grouped into performance, allocation and fund-health perspectives.
- Results are presented through an interactive Dash web application.

High-level flow

Raw Data → Data Processing → Analytical Calculations → Visualization → Interactive Decision Support

3. Analytical Coverage

- Current holdings and portfolio value
- Profit/loss and return analysis
- Fund-level and portfolio-level XIRR
- Historical purchase and redemption analysis
- Asset and market-cap allocation
- Fund comparison and portfolio concentration

4. Dashboard Experience

The application is organized around practical analytical questions rather than simply displaying raw data.

Portfolio Overview

Provides a high-level view of portfolio value, investment, profit, return and overall allocation.

Holdings

Shows current fund-level holdings with interactive sorting, filtering and pagination.

Performance

Brings together portfolio and fund performance, investment/redemption trends and contribution analysis.

Fund Health

Provides comparative fund-level views to identify stronger and weaker performers and areas requiring attention.

5. Technology & Tools

The dashboard was built using Python-based data processing and analytics, with Pandas and NumPy for data preparation, SciPy for financial calculations, Plotly for interactive visualizations and Dash for the web application layer. Excel/OpenPyXL is used as the input-data interface.

The project is packaged with dependency documentation, maintained in GitHub and deployed as a live web application.

6. Performance & XIRR

A key feature is the use of dated cash flows to evaluate investment performance rather than relying only on simple point-to-point returns. This allows the dashboard to account for the timing of purchases, redemptions and the current portfolio value.

The resulting XIRR view provides a more meaningful comparison of performance across funds with different transaction histories.

7. From Descriptive to Advanced Analytics

The current dashboard primarily provides descriptive and diagnostic analytics: what has happened, where the portfolio stands and which areas deserve attention.

Potential next-stage capabilities include:

- Predictive forecasting of portfolio performance and cash flows
- Risk and drawdown analysis
- Scenario and what-if modelling
- Benchmark and peer comparison
- Anomaly and early-warning detection
- Rule-based or model-driven recommendations
- Automated data refresh through APIs or databases
- Machine-learning models for forecasting and classification

8. Reusable Business Analytics Framework

Although demonstrated using mutual fund data, the same design pattern can support many business functions.

Domain	Possible dashboard application
Finance	Portfolio, revenue, profitability and financial performance
WFM / Operations	Capacity, utilization, staffing and service performance
Sales	Revenue, pipeline, conversion and target tracking
Customer Analytics	Customer behaviour, service levels and retention
Project Analytics	Delivery, resource utilization, cost and milestone performance
Management BI	KPI monitoring, trend analysis and decision support

The core transferable skill is not the mutual-fund domain itself; it is the ability to translate a business question into data preparation, analytical logic and an intuitive decision-support interface.

9. Project Delivery

The project is structured as a complete portfolio-ready analytics solution rather than a standalone script.

- Anonymized demonstration dataset
- Interactive web dashboard
- Source code and dependency documentation
- Project case-study documentation
- GitHub repository for versioned source and assets
- Public Render deployment for browser-based demonstration

10. What the Project Demonstrates

- Data transformation from raw transactional records to analytical datasets
- Business-oriented KPI and performance analysis
- Financial and time-based return calculations
- Interactive visualization and dashboard design
- Reusable analytics architecture across business domains
- End-to-end delivery from data source to live application

Project Links

[Live Dashboard](#)

[GitHub Repository](#)

The mutual fund dashboard is the demonstration use case; the broader objective is to build scalable, business-focused analytics dashboards that can be adapted to different datasets and decision-making environments.